Guopeng Liang, Ph.D.

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PROFESSIONAL EXPERIENCE

Assistant Professor, University of Hong Kong (2026 – present)
Brown Postdoctoral Fellow, Yale University (2024 – 2026; supervisor: Michelle Wong)
Postdoctoral Associate, University of Minnesota (2022 – 2024; supervisor: Peter Reich)

EDUCATION

Ph.D., Utah State University (2022; supervisor: Bonnie Waring and John Stark)
M.S., Chinese Academy of Agricultural Sciences (2016)
B.S., Shandong Agricultural University (2013)

RESEARCH INTERESTS

My research investigates how global change factors (e.g., warming and drought), agricultural management (e.g., fertilization and tillage), and biodiversity shape soil carbon cycling and plant productivity. By integrating field experiments, laboratory incubations, molecular tools, stable isotopes, meta-analyses, and biogeochemical modeling, my long-term research program seeks to advance nature-based solutions for carbon sequestration, climate mitigation, and food security.

SELECTED PUBLICATIONS (12 out of 80+)

*H-index = 33, Citations > 3,800 (Source: Google Scholar; *: corresponding author).*

1. **Liang, G***., Stefanski, A., Eddy, W., Bermudez, R., Montgomery, R., Hobbie, S., Rich, R., Reich, P* (2024) Soil respiration response to decade-long warming modulated by soil moisture in a boreal forest. ***Nature Geoscience***.
2. **Liang, G**., Stark, J., Waring, B (2023) Mineral reactivity determines root effects on soil organic carbon. ***Nature Communications***.
3. **Liang, G***., Schlesinger, W (2026) Potassium fertilization enhances both cereal yield and soil organic carbon: a meta-analysis. ***Nature Communications***.
4. **Liang, G*** (2025) Tree diversity increases soil carbon accumulation. ***Nature Reviews Biodiversity***.
5. **Liang, G***., Sun, P., Waring, B., Fu, Z., Reich, P (2025) Alleviating nitrogen and phosphorus limitation does not amplify potassium-induced increase in terrestrial biomass. ***Global Change Biology***.
6. **Liang, G***., Luo, Y., Zhou, Z., Waring, B (2021) Nitrogen effects on plant productivity change at decadal time-scales. ***Global Ecology and Biogeography***.
7. **Liang, G*** (2022) Nitrogen fertilization mitigates global food insecurity by increasing cereal yield and its stability. ***Global Food Security***.
8. **Liang, G*** (2025) Global pattern of warming effects on microbial respiration is explained by soil microbial biomass carbon and nitrogen. ***CATENA***.
9. **Liang, G**., Reed, S., Stark, J., Waring, B (2023) Unraveling mechanisms underlying effects of wetting-drying cycles on soil respiration in a dryland. ***Biogeochemistry***.
10. Isbell, F. et al., incl. **Liang, G.** (2026) Predicting temporal stability and resilience from resistance and recovery. ***Nature***.

11. Tang, S. et al., incl. **Liang, G**. (2025) Soil carbon sequestration enhanced by long-term nitrogen and phosphorus fertilization. ***Nature Geoscience***.
12. Zhou, T., **Liang, G**., Reich, P., Delgado-Baquerizo, M., Wang, C., Zhou, Z (2024) Promoting effect of plant diversity on soil microbial functionality is amplified over time. ***One Earth***.

ACADEMIC SERVICE

Associate Editor: *Global Ecology and Biogeography*, *Carbon Balance and Management*, and *Soil Use and Management*

Subject-matter Editor: *Ecological Monographs* and *Plant and Soil*

Editorial Board Member: *Communications Earth and Environment*, *Global Change Biology*, and *New Phytologist*

Grant Review Panelist: USDA (2024) and NSF grants (2026)

Committee Chair: AGU Biogeosciences Section Fall Meeting Planning Committee (2026)

SELECTED HONORS & AWARDS

- **Stanford/Elsevier's Top 2% Scientist**, 2024-2026
- **Exchange Professor Fellow** (University of Michigan), 2026
- **Synthesis Skills for Early Career Researchers Fellow** (National Center for Ecological Analysis and Synthesis, USA), 2024-2025
- **Environmental Leadership and Mentoring Certificate Program Fellow** (Yale School of the Environment), 2024-2025
- **U.S. Carbon Program Leadership Award** (NASA, NOAA, USDA, and USFS), 2023
- **Gene E. Likens Award** (Biogeosciences Section of the Ecological Society of America), 2024
- **Biogeosciences Leaders of Tomorrow** (*JGR-Biogeosciences*), 2025
- **Early Career Award** (Asian Ecology Section of the Ecological Society of America), 2025
- **Sino-Eco Young Investigator Paper Award** (Sino-Ecologists Association Overseas), 2024

TEACHING EXPERIENCE

- Teaching Assistant, Biology lab. Utah State University, 2019.
- Teaching Assistant, Microbiology lab. Utah State University, 2019.

MENTORING EXPERIENCE

- Yanjie Zhu (2026-present), Postdoc at the University of Hong Kong.
- Wei Ke (2026-present), PhD student at the University of Hong Kong.
- Yundi Bai (2020-2021), master's student at Imperial College London.
- Camilla Moses (2020-2021), an undergraduate technician at Utah State University.
- Jalynn Jones (2019-2021), an undergraduate technician at Utah State University.
- Preston Christensen (2018-2019), an undergraduate technician at Utah State University.

GRANTS

- Soil carbon dynamics under global change. Funded by Cedar Creek Long-term Ecological Research Site, 2024, Co-PI, $20,000.
- Modeled carbon cycle responses to altered precipitation and interannual variation in desert grasslands. Funded by Sevilleta Long-term Ecological Research Site, 2018, PI, $4,000.